Gobinath Kulathunga Ashwinthan

Electrical and Electronic Engineering Graduate — RTL Design & Verification

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INTRODUCTION

Electrical and Electronic Engineering graduate focused on digital hardware design for FPGAs. Completed an industry internship at Thakshana Technologies, with hands-on experience across communications systems, signal processing hardware, and automated design verification, taking designs from specification through simulation to working FPGA implementation. Seeking an entry-level role in RTL design and verification, with a strong interest in computer architecture and processor verification. Fluent in English, Tamil, and Sinhala.

TECHNICAL SKILLS

Programming: MATLAB, Python, ARM Assembly, Verilog HDL, VHDL
Libraries Used: NumPy, SciPy, Pandas, Matplotlib, TensorFlow, PyTorch, OpenCV, Scikit-learn
Microcontrollers/Boards: ESP32, Arduino (AVR), Artix-7 (Basys-3), Zynq UltraScale+ MPSoC (XCZU9EG)
Tools: Arduino IDE, Proteus, KiCad, Vivado Xilinx, PS-CAD

PROJECTS

MIL-STD-188-110B HF Modulator — FPGA RTL Design & Implementation

May - Jun 2026

VHDL, Vivado, MATLAB, AXI4-Stream, Avalon-ST, ILA, Zynq UltraScale+ ZCU102

Overview: Designed and implemented a **MIL-STD-188-110B-compliant HF modem transmit chain** in synthesizable **VHDL**, supporting multiple data rates (**150–1200 bps**) and interleaving configurations.

Contributions: Developed key DSP and baseband modules including **CDC** interfaces, 1/2 rate convolutional **FEC encoder**, block **interleaver**, **known-data insertion**, and **8-PSK symbol mapper**, with a parameterized RTL architecture and efficient dataflow control. Performed MATLAB-based functional verification, RTL simulation, and timing optimization, and integrated the design on the **AMD Xilinx Zynq UltraScale+ ZCU102** at **500 MHz** using **AXI4-Stream** and **Avalon-ST** interfaces and on-chip logic analyzers (**ILA**) with JTAG.

Fixed-Point Standard Deviation IP — FPGA DSP Architecture & Optimization

Jul - Aug 2026

VHDL, MATLAB, Vivado, Xilinx CORDIC IP, BRAM LUTs, DSP Slices, Zynq UltraScale+

Overview: Designed and implemented a **parameterized, synthesizable VHDL IP core** for **fixed-point standard deviation** computation, targeting FPGA-based digital signal processing applications. Built a resource-efficient **two-pass architecture** covering fixed-point addition, subtraction, multiplication, division, and **square-root computation using Xilinx CORDIC IP**, with optimized word lengths and numerical precision.

Contributions: Optimized the datapath using **BRAM-based reciprocal lookup tables**, **DSP slice utilization**, and **arithmetic restructuring**, achieving a **37.5% reduction in DSP usage** (**16 to 10 slices**). Performed **MATLAB-based fixed-point modeling**, RTL verification, and timing/resource optimization on the **AMD Xilinx Zynq UltraScale+** platform.

Automated RTL Verification Framework — FPGA Demodulator

Sep - Oct 2026

Python, Tcl, Verilog, VHDL, XSim, QuestaSim, Regression Testing

Overview: Developed a **Python-based RTL verification framework** integrating **Verilog testbenches** and **Tcl scripting** for automated **system-level** and **incremental bottom-up module verification** of an FPGA demodulator.

Contributions: Implemented **dynamic RTL configuration** using **VHDL generate constructs**, enabling flexible module integration and targeted debugging through command-line arguments. Integrated **XSim** and **QuestaSim** with configurable batch/GUI execution, reducing simulation runtime through selective compilation and direct simulator invocation. Automated **test-vector generation**, **golden-reference comparison**, output logging, and **regression testing** for **bit-accurate functional verification**.

8-bit ALU Design *Dec 2025*

Verilog HDL, Digital Arithmetic Modules

Overview: Designed a fully structural 8-bit Gate-Level ALU supporting **signed** and **unsigned** arithmetic with a **ripple-carry two's complement adder–subtractor** and explicit overflow detection and multiplication using a **sequential shift-and-add architecture** and division via an **iterative restoring division algorithm** with shared subtract–compare logic. Integrated multi-cycle control (**start/busy/done handshake**) and a complete flag set: zero, negative, overflow, carry-out, and divide-by-zero.

Parameterized Synchronous FIFO in Verilog *Feb 2026*

Verilog HDL, RTL Verification, Memory Architecture

Overview: Designed and verified a **parameterized single-clock synchronous FIFO** in Verilog RTL with configurable **depth** and **data width**. Implemented a **circular buffer** using phase-extended read and write pointers for reliable full and empty detection. Verified correct operation across **pointer wraparound** and **concurrent read/write conditions**. Developed a **task-based testbench** with directed scenarios and waveform analysis to validate data orderingand status flag behavior.

Sobel Edge Detection using Verilog *Feb 2026*

Verilog HDL, FPGA, VGA Controller

Overview: Implemented a real-time streaming Sobel edge detection pipeline on Artix-7 (Basys-3) using Verilog RTL, processing **640×480 VGA at 25 MHz**. Designed **dual-BRAM line buffers** and a **3×3 window generator** to achieve one-pixel-per-clock convolution. Integrated RGB565 image ROM, fixed-point grayscale conversion, pipelined gradient computation, saturation, and dynamic threshold control. Demonstrated multiple **display modes** (RGB, grayscale, edge, overlay) with precise VGA timing and latency alignment.

Other Projects: Subsurface emergency transmitter using near-field magnetic induction (KiCad PCB design); image compression using Huffman coding (Python).

EXPERIENCE

FPGA & DSP Intern May – Oct 2026

Thakshana Technologies *Colombo, Sri Lanka*

- Designed and implemented an HF modem transmitter on FPGA.
- Developed an automated verification framework for an FPGA demodulator.
- Developed a fixed-point standard deviation IP core for FPGA.

EDUCATION

VLSI CAD: Logic (Part I) — Ongoing Online Course Ongoing

Coursera, University of Illinois Urbana-Champaign

VHDL for FPGA Design 2026

AMD (Xilinx) Adaptive Computing Training

Designing with Vivado Design Suite 2026

AMD (Xilinx) Adaptive Computing Training

MATLAB Onramp 2026

MathWorks

B.Sc. Engineering (Hons.) in Electrical and Electronic Engineering (GPA - 3.31) 2022 – 2026

University of Peradeniya

G.C.E. Advanced Level and Ordinary Level Examinations 2012 – 2021

Royal College, Colombo 07

Primary Education 2007 – 2011

Colombo Hindu College, Bambalapitiya

EXTRACURRICULAR ACTIVITIES

Karate - University of Peradeniya 2023

Sri Lanka University Games (SLUG 2023)

- Represented University of Peradeniya in Karate at the Sri Lanka University Games 2023, showcasing discipline, teamwork, and competitive spirit.

Diploma in Electronic Keyboard 2025 – Present

London College of Music (LCM), University of West London

- Pursuing a diploma in Electronic Keyboard through the London College of Music, refining musical expression and technical proficiency.

REFERENCES

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